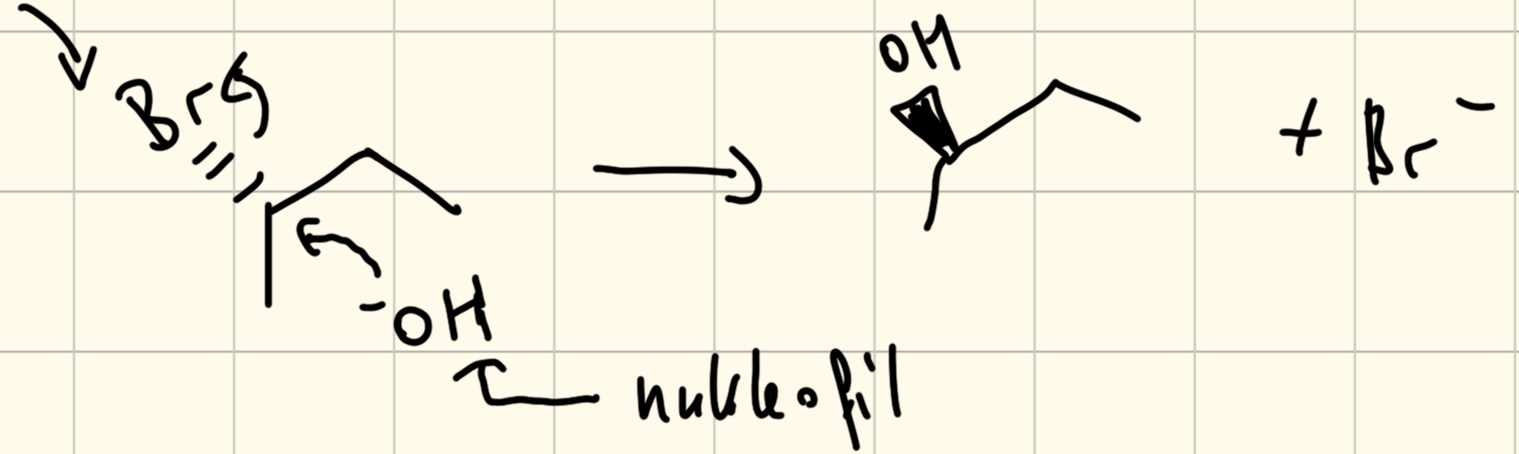
1)

$$v \propto \underbrace{[OH^-] \times [\text{substrat}]}_{SN2}$$

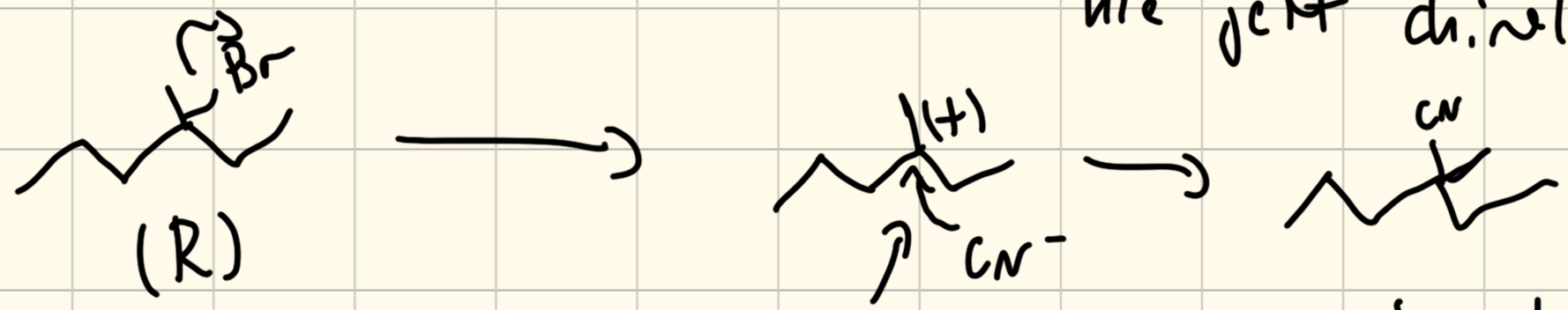
grupa opuszczająca



Z substratu o konfiguracji S powstaje produkt R, inwersja konfiguracji = SN2

2 substratu R → produkty R + S = SN1

(produkt pośredni, karbokation nie jest chiralny

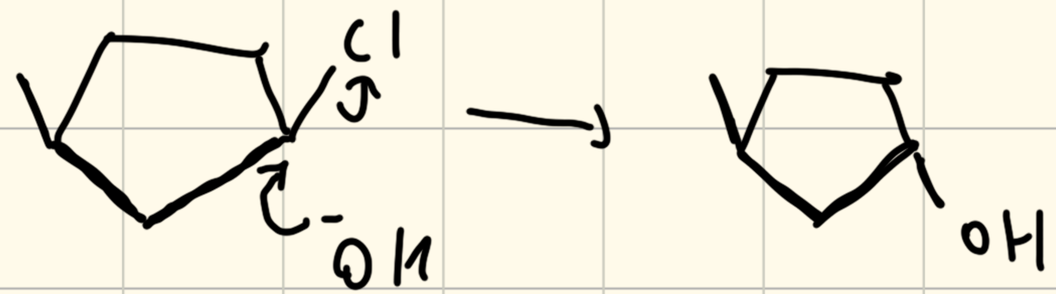


3° stabilizowany karbokation

mieszanina racemiczna

SN1 prowadzi do utraty konfiguracji (racemizacja) bo karbokation jest płaski (sp^2):

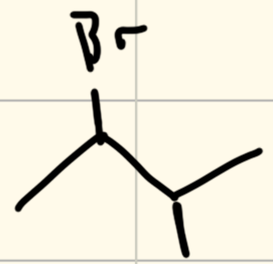
3



SN2 = inwersja konfiguracji.

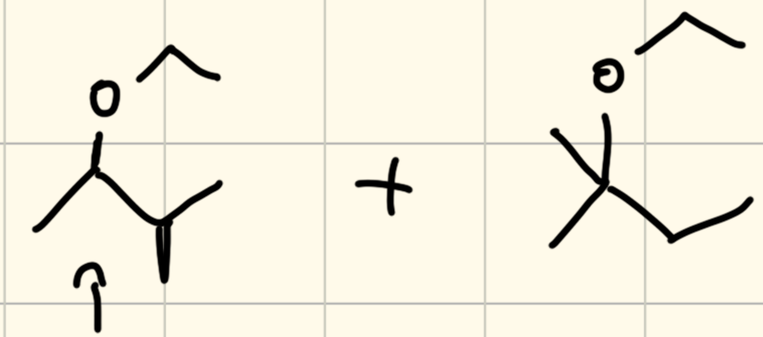
4

substrat



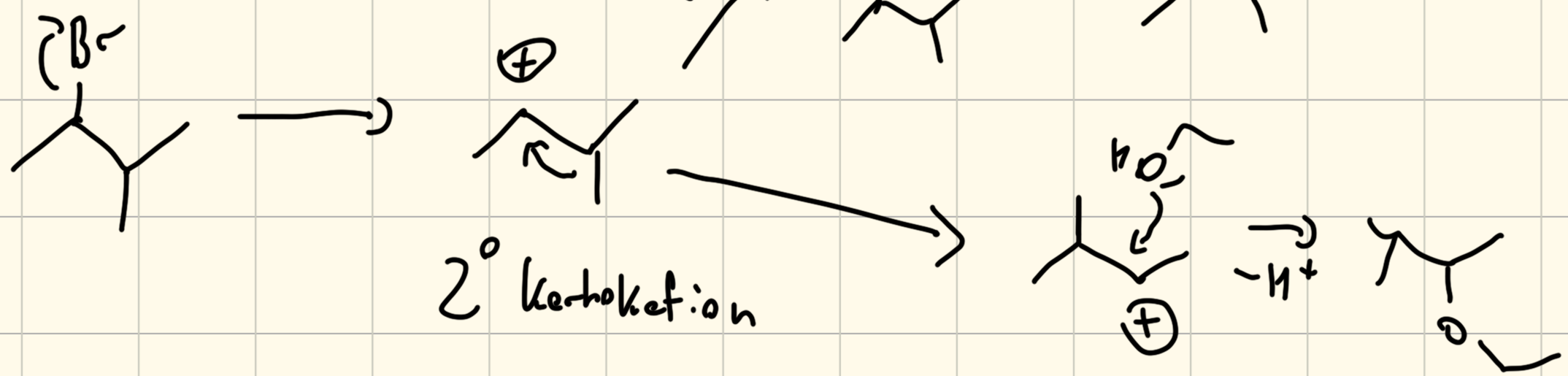
produkty:

2-etoksy-2-metylobuten

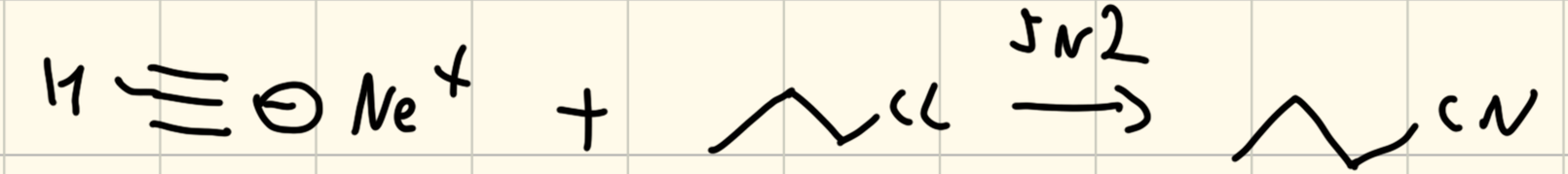


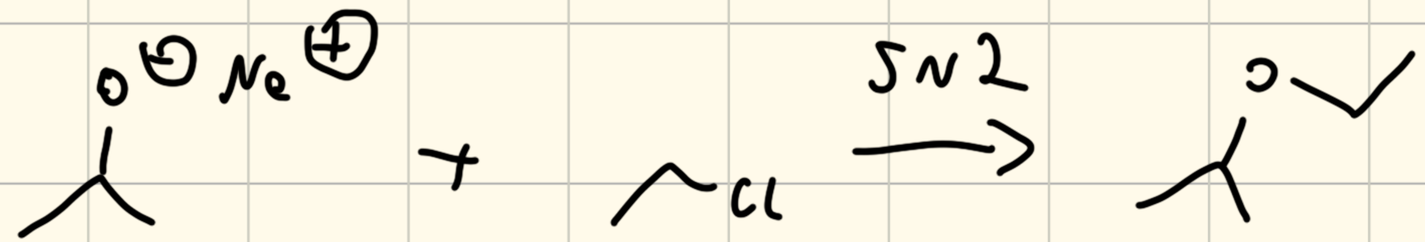
2-etoksy-3metylobuten

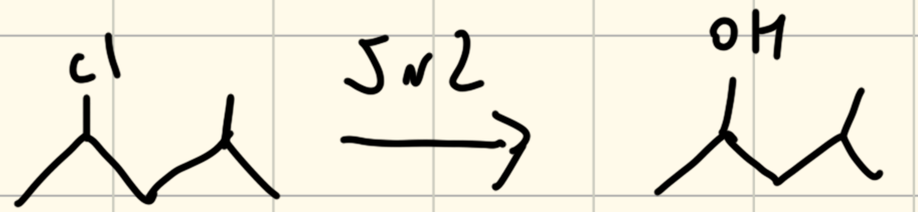
Winna przegrupowanie karbokationu



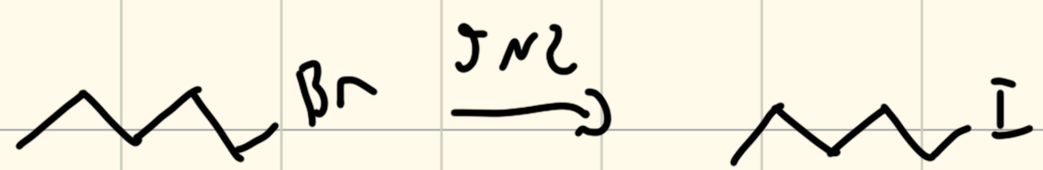
5)

a) 

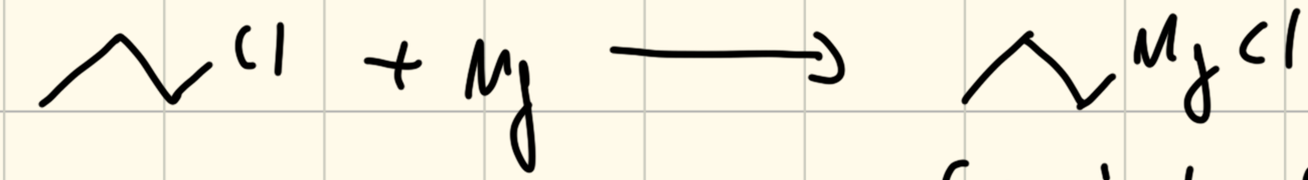
b) 

c) 

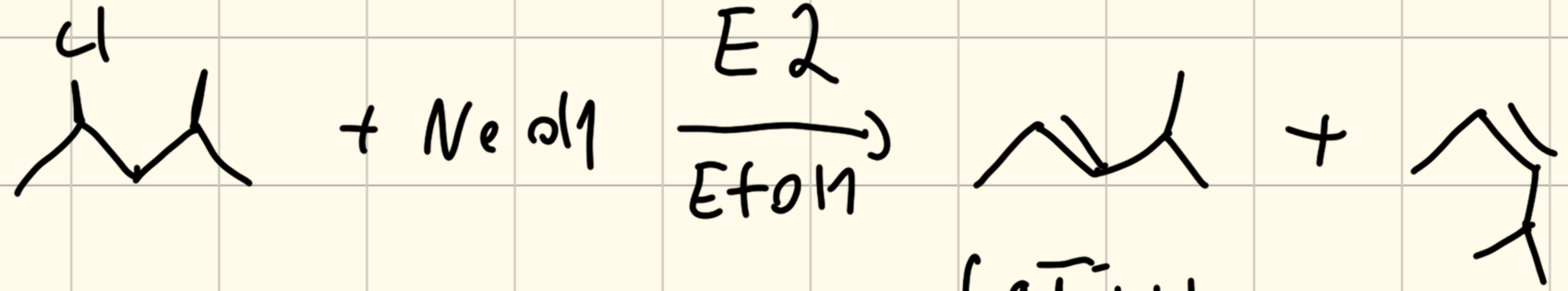
(jak podgrzejemy, może być też E2)

d) 

(śleby zachodzi)

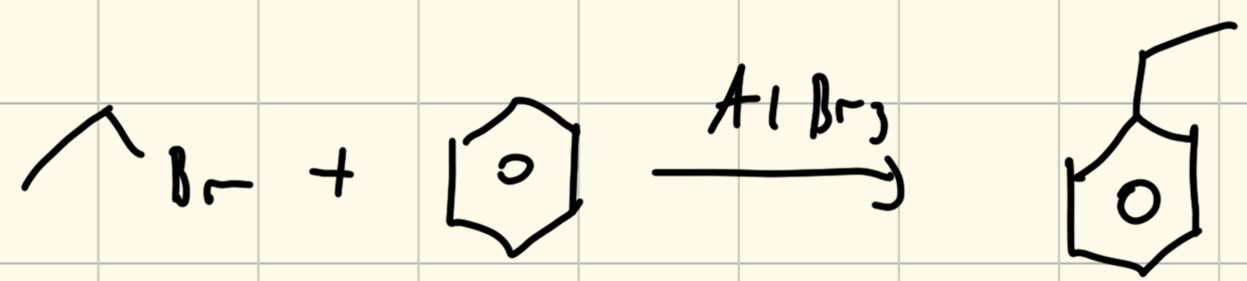
f) 

(związek Grignarda)

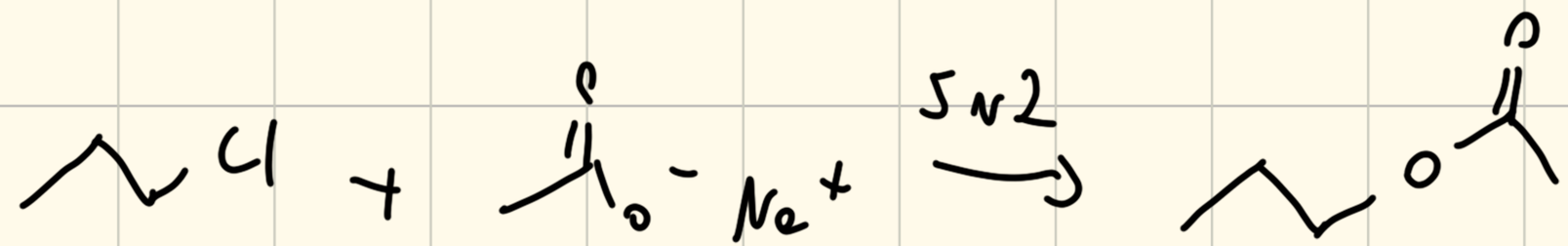
g) 

(główny produkt to trans-izomer)

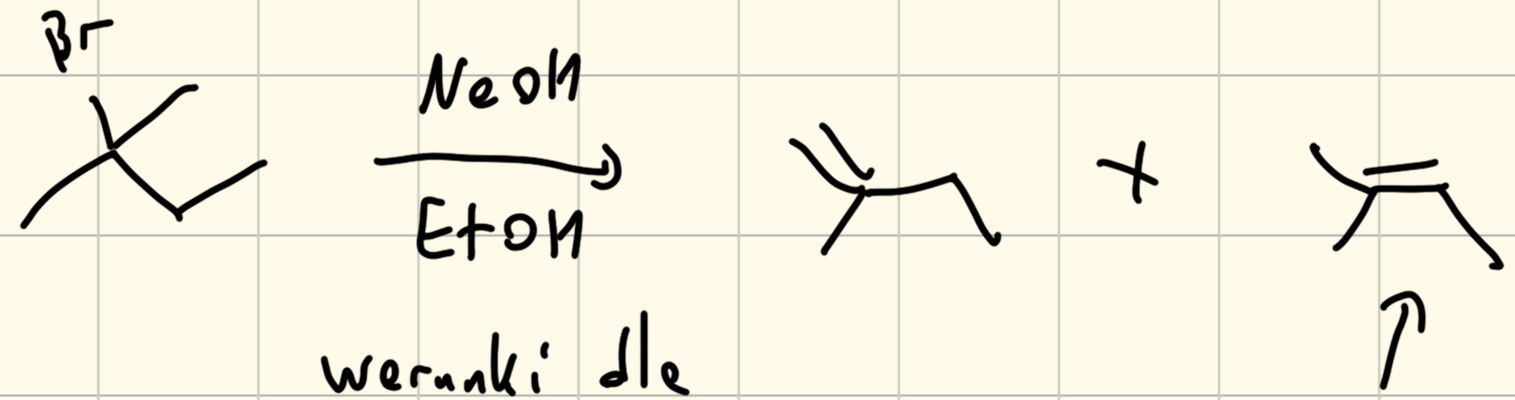
h)



i)



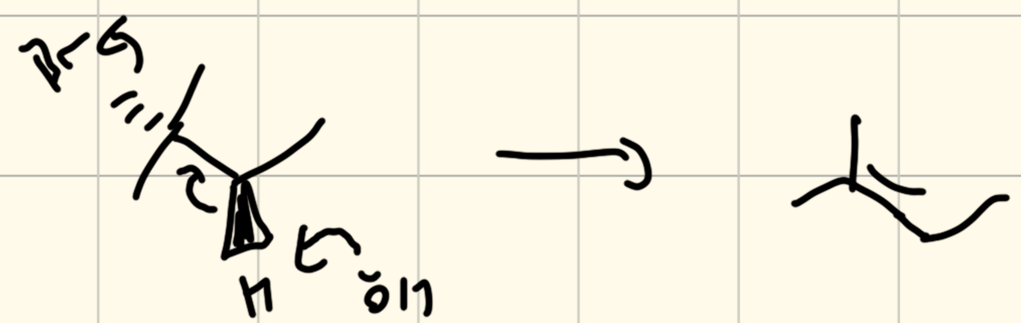
(6)



warunki dla eliminacji E2

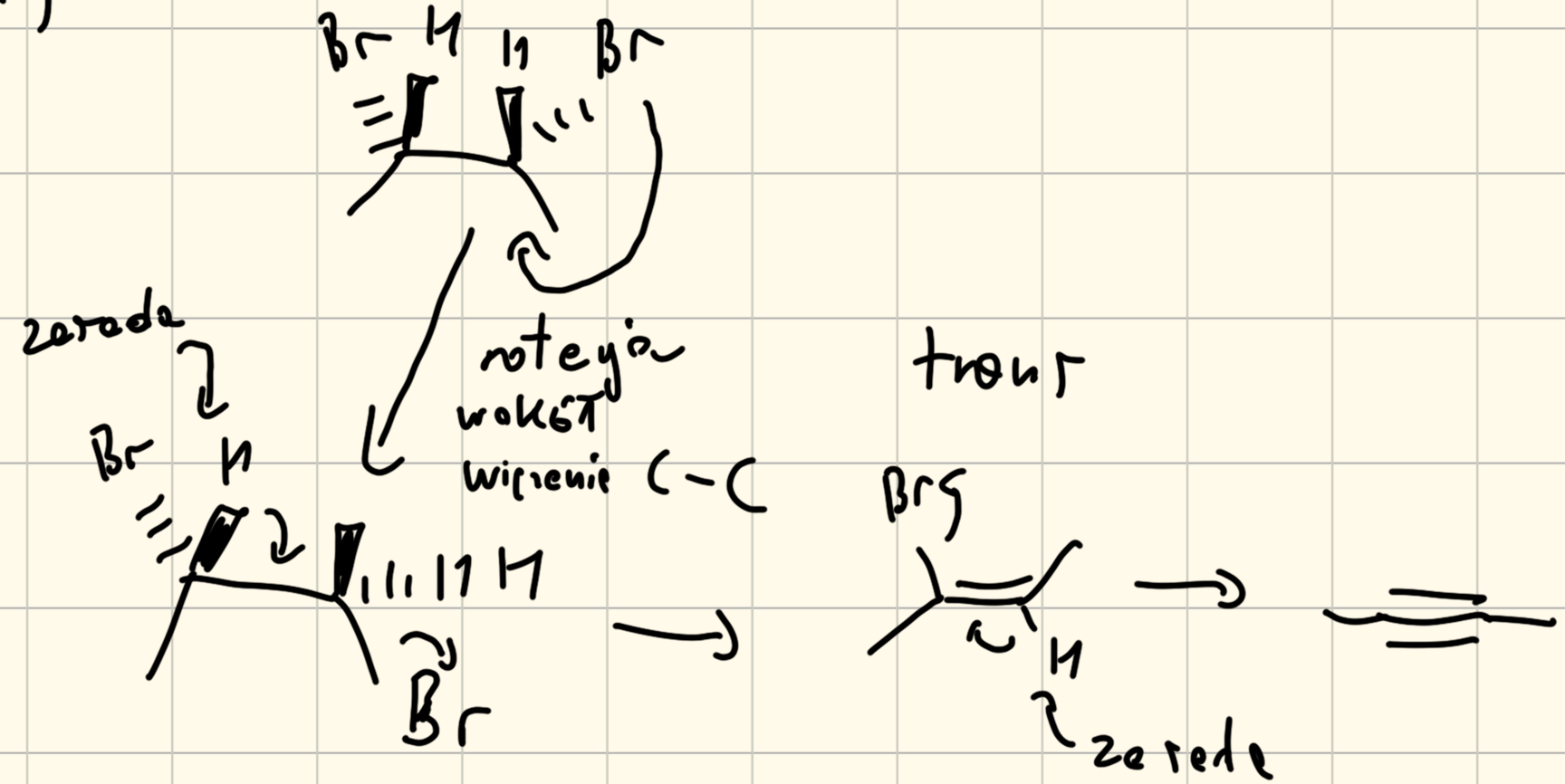
produkt główny, bardziej podstawiony

Silny nukleofil w etanolu = E2

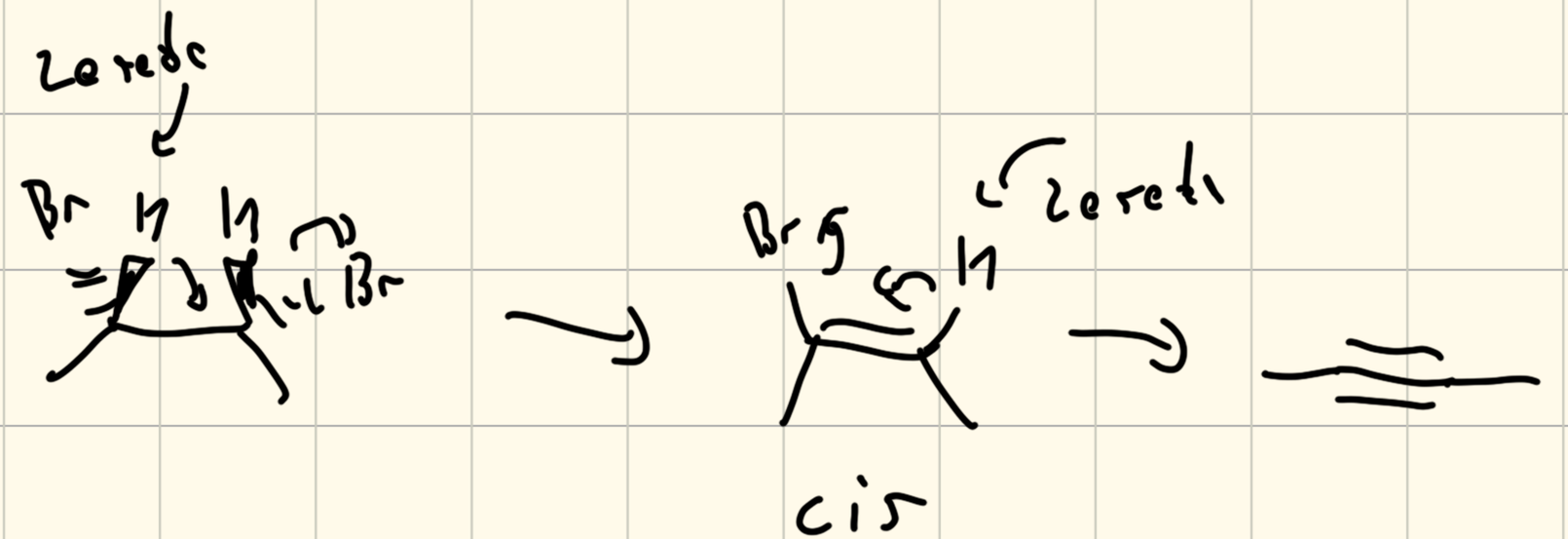


7)

a)

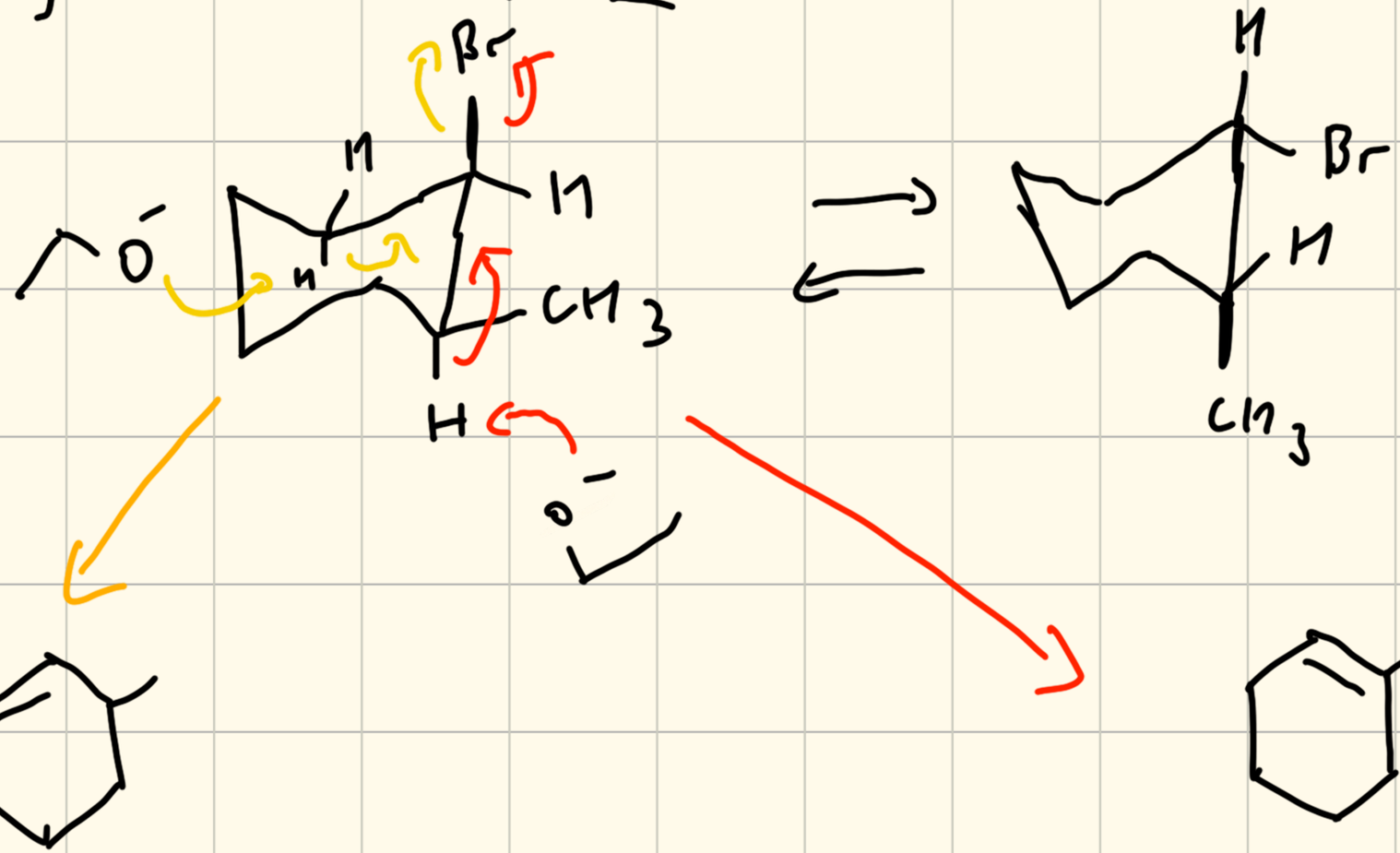

Br H H Br
zasada
rotacja wokół wiązania C–C
Br H
H H H
Br
trans
Br
H
zasada

b)


zasada
Br H H
Br
Br
H
zasada
cis

8)

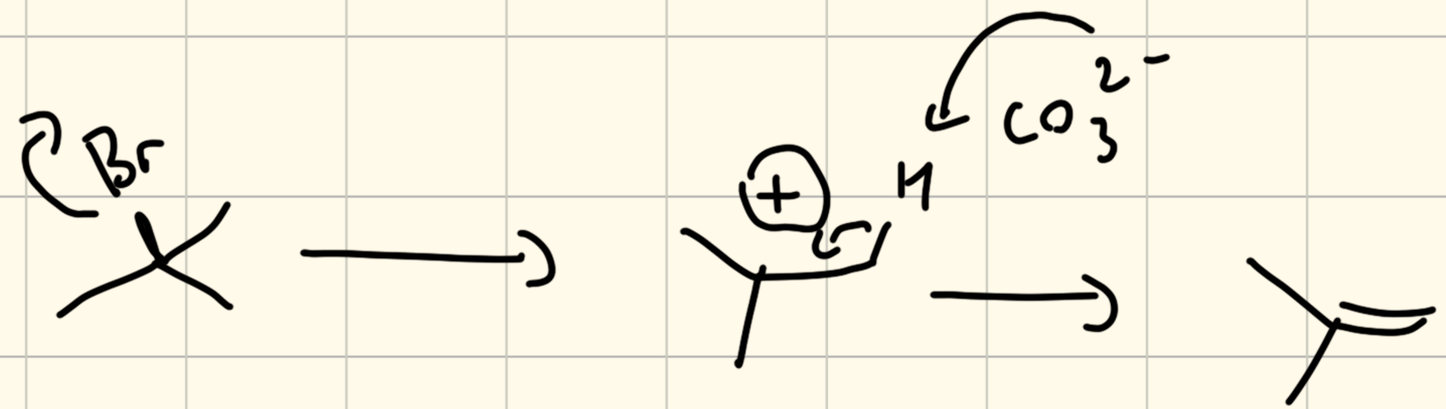
cis 1-2 →

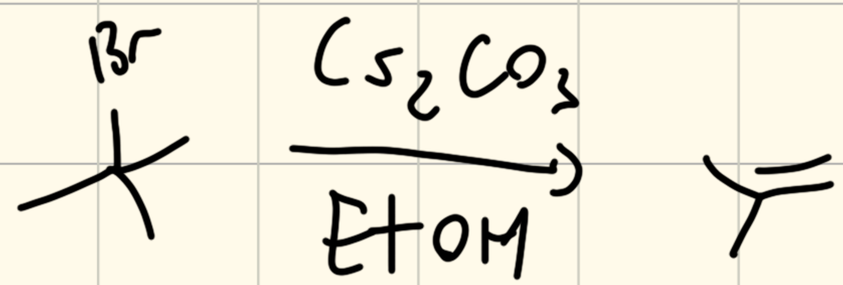


główny produkt

(zgodny z Zajcewem)

9) E1 = można użyć 3° halogenek i słabą zasadę





Zasada powinna być dość słaba dla E1 (proponuję Cs_2CO_3)